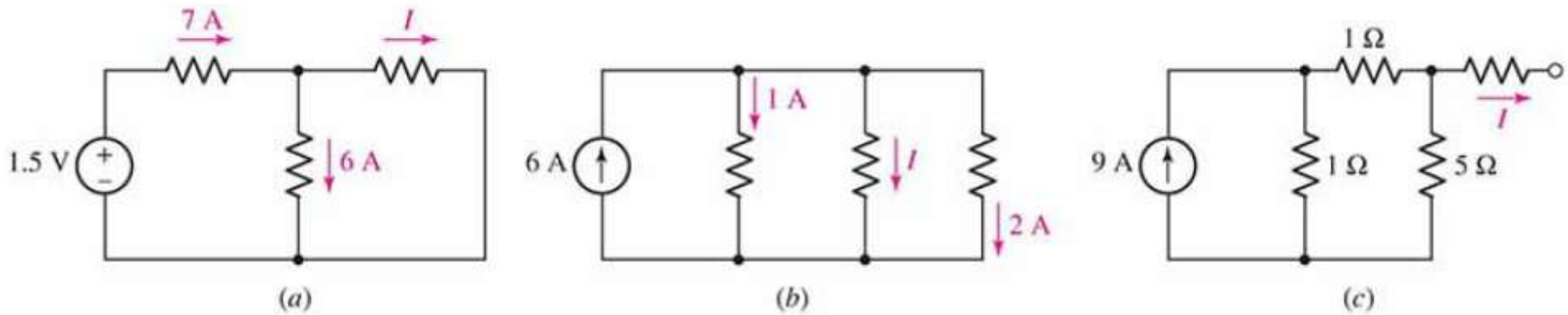
1. Use Kirchhoff's Current Law and Kirchhoff's Voltage Law to solve the following questions.

(1). Determine the current labeled I in each of the circuits below.[15]



63. Consider the seven-element circuit depicted in Fig. 3.99. (*a*) How many nodes, loops, and branches does it contain? (*b*) Calculate the current flowing through each resistor. (*c*) Determine the voltage across the current source, assuming the top terminal is the positive reference terminal.

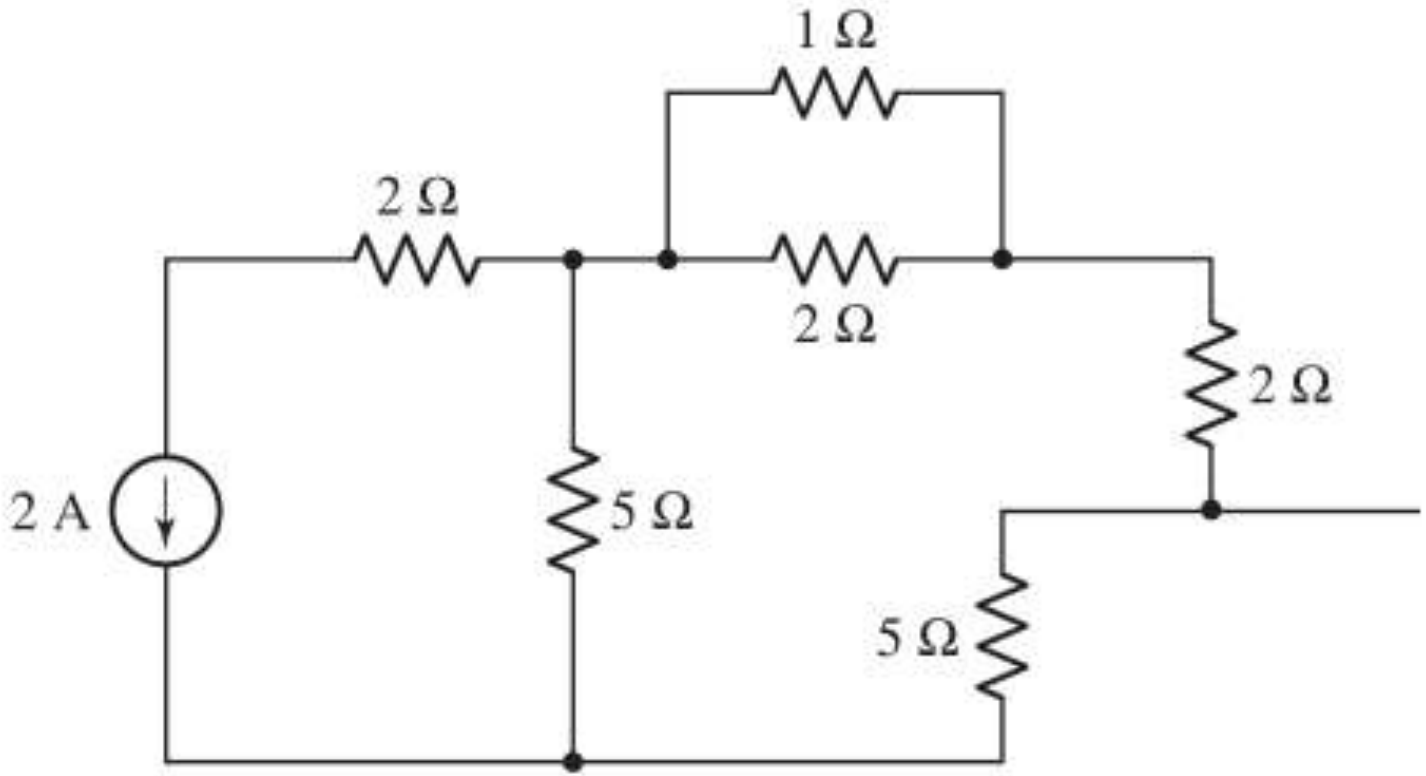


FIGURE 3.99

24. The circuit shown in Fig. 3.65 includes a device known as an op amp. This device has two unusual properties in the circuit shown: (1) $V_d = 0$ V, and (2) no current can flow into either input terminal (marked "−" and "+" inside the symbol), but it *can* flow through the output terminal (marked "OUT"). This seemingly impossible situation—in direct conflict with KCL—is a result of power leads to the device that are not included in the symbol. Based on this information, calculate V_{out}. (*Hint*: Two KVL equations are required, both involving the 5 V source.)

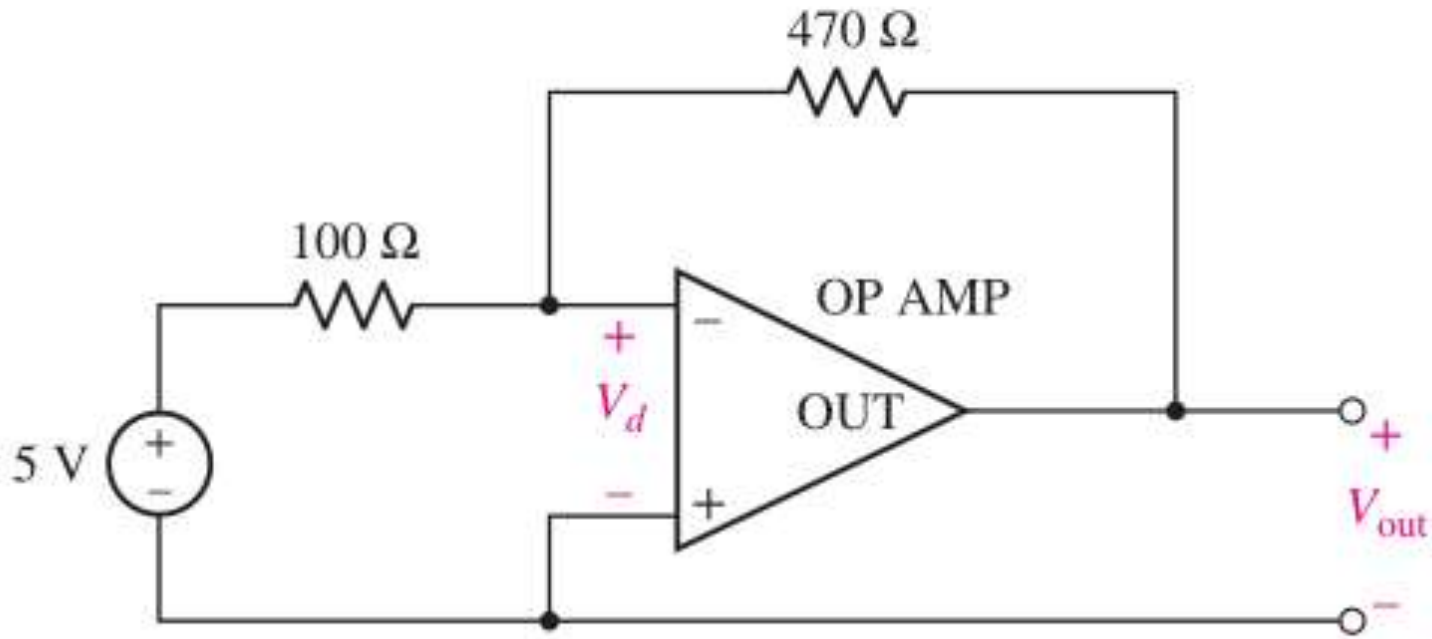


FIGURE 3.65